

The Grammar of Quantity

Units as coordinates, constants as vectors, & school algebra in service of physics

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Abstract

A unit, looked at properly, is a set of coordinates. Once we agree to track only the base dimensions of mass, length, time and temperature, every physical quantity is fixed by the powers to which those dimensions are raised, so that a velocity sits at $(0, 1, -1, 0)$ and an energy at $(1, 2, -2, 0)$, each a vector in the same space; and building one quantity as a product of powers of others is, in this language, the taking of a linear combination of such vectors. To ask which combination of given constants carries a desired dimension is therefore to solve a system of linear equations, and the character of that system, whether it has one solution, too many, or too few, mirrors the physics with uncommon faithfulness. This essay follows that single thread through three fundamental quantities of space, the Planck length with its area and volume, the mean density of the universe, and the entropy of a black hole, using each to introduce one further rung of a ladder that climbs from the back-substitution a pupil meets first, through to Gaussian elimination.

1 Dimension as a coordinate

The constants we shall need carry the dimensions

$$[c] = \text{L T}^{-1}, \quad [h] = \text{M L}^2 \text{T}^{-1}, \quad [G] = \text{M}^{-1} \text{L}^3 \text{T}^{-2}, \quad [k_B] = \text{M L}^2 \text{T}^{-2} \Theta^{-1},$$

together with the Hubble constant $[H_0] = \text{T}^{-1}$ and a plain area $[A] = \text{L}^2$. Reading each as a column of powers turns the search for a formula into a matrix equation. If we seek a product $Q = \prod_i X_i^{p_i}$ with a prescribed dimension, then collecting the exponent of each base dimension gives one linear equation per dimension, and the whole problem becomes

$$\mathbf{D} \mathbf{p} = \mathbf{t},$$

where the columns of $\mathbf{D}$ are the dimension vectors of the chosen quantities, $\mathbf{p}$ is the vector of unknown exponents, and $\mathbf{t}$ is the target dimension. The spatial part of this picture is especially clean, since the lengths, areas and volumes a student cares about lie along a single axis: asking for L^n rather than L replaces $\mathbf{t}$ by $n \mathbf{t}$, and because the map is linear the exponents simply scale by n . Whatever recipe builds a fundamental length builds its area and volume by doubling and tripling the same powers.

What remains is to solve $\mathbf{D} \mathbf{p} = \mathbf{t}$, and a linear system can behave in exactly three ways. It can have a single solution, it can be overdetermined yet consistent, and it can be underdetermined, leaving a family of solutions to be narrowed by some condition from outside. The three quantities below were chosen because each lands squarely in one of these cases.

2 One solution: a fundamental length

The Planck length is the unique combination of c , h and G with the dimension of length, written $\ell_P = c^\alpha h^\beta G^\gamma$ up to a pure number. Matching the dimensions M, L, T gives the system

$$\begin{aligned} M : \quad & \beta - \gamma = 0, \\ L : \quad & \alpha + 2\beta + 3\gamma = 1, \\ T : \quad & -\alpha - \beta - 2\gamma = 0. \end{aligned}$$

Back-substitution. The first method a pupil owns is to read a relation off one equation and feed it forward. The mass line gives $\beta = \gamma$ at once; the time line then reads $\alpha = -\beta - 2\gamma = -3\beta$; and the length line becomes $-3\beta + 2\beta + 3\beta = 2\beta = 1$, so that $\beta = \frac{1}{2}$, whence $\gamma = \frac{1}{2}$ and $\alpha = -\frac{3}{2}$. It is quick here because the substitutions happen to fall in a convenient order, and that is also its weakness: the method offers no guarantee that such an order exists, and with more constants the bookkeeping soon outgrows the page.

Gaussian elimination, and the pivot. The substitution just done succeeded because the equations could be untangled in a lucky order, and the purpose of elimination is to manufacture that order on demand.

We first set the system out as an *augmented matrix*, a row for each equation and a column for each unknown, with the right-hand sides written after a vertical bar, and we work on whole rows at a time.

$$\left[\begin{array}{ccc|c} 0 & 1 & -1 & 0 \\ 1 & 2 & 3 & 1 \\ -1 & -1 & -2 & 0 \end{array} \right],$$

To *pivot* on an entry is then to use the row it lies in to remove that unknown from every other row, subtracting from each the multiple of the pivot row that cancels the entry standing in the pivot's column; the entry put to this use is the pivot, and forming that multiple means dividing by it, so a pivot must be a number one may divide by. Sweeping down the columns in this way empties everything below the leading diagonal and leaves each row holding one unknown fewer than the row above it, which is the staircase that back-substitution needs. We thus begin with the first column, meaning to pivot on the top-left entry and clear the column beneath it; but that entry is a zero,

$$\left[\begin{array}{ccc|c} \boxed{0} & 1 & -1 & 0 \\ 1 & 2 & 3 & 1 \\ -1 & -1 & -2 & 0 \end{array} \right],$$

and clearing an entry below it would ask us to divide by that zero, which is barred. The repair is to exchange the first two rows, lifting a usable entry into the corner,

$$\left[\begin{array}{ccc|c} \boxed{1} & 2 & 3 & 1 \\ 0 & 1 & -1 & 0 \\ -1 & -1 & -2 & 0 \end{array} \right].$$

The 1 in the corner is now the pivot; adding its row to the third cancels the -1 beneath it and clears the column,

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 0 & \boxed{1} & -1 & 0 \\ 0 & 1 & 1 & 1 \end{array} \right],$$

the 1 it exposes in the second column is the next pivot, and subtracting its row from the third clears that column in turn,

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & \boxed{2} & 1 \end{array} \right],$$

leaving the 2 in the third column as the last pivot.

The system is now a staircase, and back-substitution climbs it from the bottom, dividing by one pivot at each rung to release one unknown. The last pivot gives $2\gamma = 1$, so $\gamma = \frac{1}{2}$; the middle pivot gives $\beta - \gamma = 0$, so $\beta = \frac{1}{2}$; the first gives $\alpha + 2\beta + 3\gamma = 1$, so $\alpha = -\frac{3}{2}$. The same three pivots that cleared the columns going down are the three numbers we divide by coming back up, and because every column carried one, every unknown is released in turn and the answer is a single point,

$$\ell_P = \sqrt{\frac{hG}{c^3}} \approx 1.6 \times 10^{-35} \text{ m}.$$

The spatial dimensions for free. Because the target enters only through the right-hand column, asking instead for an area or a volume multiplies that column by two or three and, by linearity, the exponents with it. The Planck area and volume therefore require no fresh calculation,

$$\ell_P^2 = \frac{hG}{c^3} \approx 2.6 \times 10^{-70} \text{ m}^2, \quad \ell_P^3 = \left(\frac{hG}{c^3}\right)^{3/2} \approx 4.2 \times 10^{-105} \text{ m}^3.$$

3 Too many equations, and consistent: the density of the cosmos

The mean density of the universe is a quantity of dimension ML^{-3} , the negative cube that any density wears, and on the largest scale it is set by the Hubble constant and the strength of gravity alone, $\rho = H_0^p G^q$. Now there are three base dimensions to satisfy and only two exponents to satisfy them with, so the system is taller than it is wide:

$$\left[\begin{array}{cc|c} 0 & -1 & 1 \\ 0 & 3 & -3 \\ -1 & -2 & 0 \end{array} \right] \longrightarrow \left[\begin{array}{cc|c} 1 & 2 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & 0 \end{array} \right].$$

The elimination places a pivot in each of the two columns and then meets a third row with no column left to pivot in; that row collapses to zeros, which is what a redundant equation looks like once the work is done, the third equation having added nothing the first two had not already fixed. Its collapse is the certificate that the chosen constants can indeed reach the target. The two pivots determine the two unknowns, $q = -1$ and $p = 2$, so that

$$\rho = \kappa \frac{H_0^2}{G}, \quad \rho_c = \frac{3H_0^2}{8\pi G} \approx 9 \times 10^{-27} \text{ kg m}^{-3},$$

amounting to some five or six hydrogen atoms in every cubic metre of space. Had that last row instead kept a nonzero constant on the right, it would read $0 = \text{something}$, an impossibility, and the elimination would have reported that no product of H_0 and G carries the density's dimension; a row left without a pivot is redundant when its constant vanishes and contradictory when it does not.

4 Too few equations: a black hole's entropy

The entropy of a black hole brings in temperature, and with it a fourth base dimension and a fifth constant. Seeking $S = k_B^a A^b c^d h^e G^f$ with the dimension $\text{ML}^2\text{T}^{-2}\Theta^{-1}$ of entropy, the system has four equations, one for each base dimension, in five unknowns. Set out as an augmented matrix, with the five constants heading the columns and the four base dimensions labelling the rows, it reads

$$\left[\begin{array}{c|ccccc|c} & k_B & A & c & h & G & [S] \\ \hline \text{M} & 1 & 0 & 0 & 1 & -1 & 1 \\ \text{L} & 2 & 2 & 1 & 2 & 3 & 2 \\ \text{T} & -2 & 0 & -1 & -1 & -2 & -2 \\ \Theta & -1 & 0 & 0 & 0 & 0 & -1 \end{array} \right],$$

where each row imposes the homogeneity of one base dimension and the entry past the bar is the power that dimension carries in entropy itself, the target the rest must reproduce.

The temperature row carries only k_B , so it is an equation in the single unknown a , the easiest pivot the problem offers, handing over $a = 1$ before anything is cleared; this is the kind of pivot, sitting in a nearly empty row, that isolates its unknown at a stroke. Reducing the remaining rows to echelon form gives

$$\left[\begin{array}{ccccc|c} 1 & 0 & 0 & 1 & -1 & 1 \\ 0 & 2 & 1 & 0 & 5 & 0 \\ 0 & 0 & -1 & 1 & -4 & 0 \\ 0 & 0 & 0 & 1 & -1 & 0 \end{array} \right],$$

and a pivot now falls in each of the first four columns, those of k_B , A , c and h , while the fifth column, the one belonging to G , receives none. An unknown whose column carries no pivot is free, so the exponent f of G may be chosen at will, and the rows above fix the rest in terms of it, read from the bottom up: the last row gives $e = f$, the third $d = -3f$, the second $b = -f$, and the first $a = 1$, this final value holding whatever f is taken to be. That single loose exponent is where the freedom lives, and it is tidiest to follow it through the area exponent, setting $k = b = -f$, in which the whole family of dimensionally valid answers reads

$$(a, b, d, e, f) = (1, k, 3k, -k, -k) = (1, 0, 0, 0, 0) + k(0, 1, 3, -1, -1).$$

The vector added by k is the heart of the matter. As a product it reads $A c^3 / h G$, and that group is dimensionless, being precisely the horizon area measured in Planck areas, A / ℓ_P^2 ; adding any multiple of it therefore changes no dimension at all. Every member of the family carries the dimension of entropy exactly, so the units cannot prefer one over another, and the freedom is an honest report that dimensional reasoning has said all it can, rather than a defect of the calculation.

What closes the gap is physics. Bekenstein argued that the entropy must be proportional to the horizon area itself, the first power and no other, and that single demand sets $b = 1$, hence $k = 1$, collapsing the family to

$$S = \kappa \frac{k_B A c^3}{h G}, \quad \kappa = \frac{\pi}{2},$$

the only dimensionless number, κ , still standing beyond the reach of any argument about units. The progression has run its course: where the length was determined and the density overdetermined, the entropy is underdetermined, and the null direction along which it is free is the count of Planck tiles that a horizon of area A would take to cover, which is the very quantity the entropy turns out to measure.

5 The ladder, and a word to mathematics

Read in sequence the three quantities are a tour of what a linear system can do, and the methods they call for are the rungs by which a student learns to handle such systems: back-substitution while the equations stay small and obliging, Gaussian elimination once order can no longer be trusted, and the reading of pivots into rank and freedom when the answer is a family rather than a point.

The appeal, then, is not merely to physics teachers. It is an appeal to mathematics departments in high schools to recover one of mathematics' oldest and most useful offices: to serve the physical sciences by making the language of quantity precise before physics has to clean up the confusion. Too often the algebra of symbols is taught as though the symbols floated free of units, prefixes and reciprocal powers. A pupil may learn perfectly well that $(2c)^3 = 8c^3$, and yet be left unsure why

$$2\text{ cm}^3 = 2(\text{cm})^3 = 2(10^{-2}\text{m})^3 = 2 \times 10^{-6}\text{ m}^3,$$

rather than $8 \times 10^{-6}\text{ m}^3$ or $2 \times 10^{-3}\text{ m}^3$. Likewise, pupils meet negative powers in algebra, but the physical meaning of those powers is often left stranded. Reciprocal units need the same care. $[\text{N}, \text{m}^{-2} = \text{N}(\text{m})^{-2}]$ means newtons per square metre: pressure. $[\text{N}, \text{kg}^{-1} = \text{N}(\text{kg})^{-1}]$ means newtons per kilogram: gravitational field strength.

This is where mathematics can be the better servant to the sciences. It can teach powers, indices, reciprocals, simultaneous equations and vectors in a way that does not sever them from scale. Prefixes are powers of ten. Areas and volumes are powers of length. Densities, pressures and field strengths are coordinates in a dimensional space. Once that is made explicit, dimensional analysis is linear algebra with units attached.

That is why these examples belong in the mathematics classroom as well as in the physics laboratory. The Planck length, the density of the cosmos and the entropy of a black hole are grand objects, but the machinery by which their dimensions are read is ordinary school mathematics: powers, products, substitution, elimination and the geometry of solution spaces. Physics supplies the weight. Mathematics supplies the grammar. The accompanying student investigations make this route explicit. The Planck-length task asks pupils to build ℓ_P from c , h and G using dimensional homogeneity and first-difference reasoning on a lattice.¹ The black-hole entropy task then takes the same method to its limit, showing why the units determine most of the formula

$$S = \kappa \frac{k_B A c^3}{h G}$$

but cannot determine the dimensionless constant or the physical demand that entropy scale with area.² Together they make the departmental point. Dimensional analysis is not a physics trick. It is school algebra, made honest by units.

Colophon. Drafted with the assistance of a large language model, Anthropic's Claude, used in the manner one might cite a computer algebra system, as an instrument working under the author's direction and argument.

¹Lee McCulloch-James, *Building the Planck Length: a smallest ruler for space, found on a lattice*, Year 11 Physics and Mathematics task: <https://leemacjames.github.io/planckLengthDerivation.pdf>.

²Lee McCulloch-James, *Deriving Black Hole Entropy from the Homogeneity of Units*, Year 11 Physics and Mathematics task: <https://leemacjames.github.io/BlackHoleEntropy.pdf>.